Bayesian Networks for
Early Detection of Coronary Artery Disease

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Problem & Datasets

The Problem

Coronary Artery Disease (CAD) is one of the leading causes of death worldwide, yet its diagnosis often relies on costly and invasive procedures such as angiography and echocardiography.

Goal: estimate the probability of CAD from routine clinical measurements using Bayesian Networks as an interpretable computational alternative.

Cleveland Heart Disease Dataset

- 303 patients ů 13 clinical features
- Binary target: `condition` (0/1)
- Main dataset for training & evaluation
- Continuous variables discretized using clinical thresholds

Indian Heart Disease Dataset

- 765 patients ů same feature structure
- Different demographic population
- Used for **cross population generalization** test

Approach: Why Bayesian Networks & Data Discretization

Why Bayesian Networks?

- Naturally encode **causal relationships** between variables
- **Interpretable** structure: clinicians can validate edges

Clinical Discretization of continuous variables:

Cholesterol	Blood Pressure	ST Depression	Age
< 200 Desirable 200–239 Borderline ≥ 240 High NCEP ATP III	< 120 Normal 120–139 Elevated/Stage 1 ≥ 140 Stage 2+ AHA 2017	0 Optimal < 1.0 Low Risk 1.0–1.9 Moderate ≥ 2.0 High Risk Yap et al. 2005	20–29 Early adult 30–39 Early middle ... ≥ 70 Late onset GBD Study

Progressive Network Design

1

Unconstrained

Hill Climbing + AIC scoring. No constraints on structure. Purely data-driven baseline.

2

Blacklist

Forbidden edges prevent clinically implausible directions, e.g.: `condition`→`age`, `condition`→`sex`.

3

Forced Edges

Medically grounded edges added based on peer-reviewed literature, e.g. `age`→`condition`, `oldpeak`→`condition`.

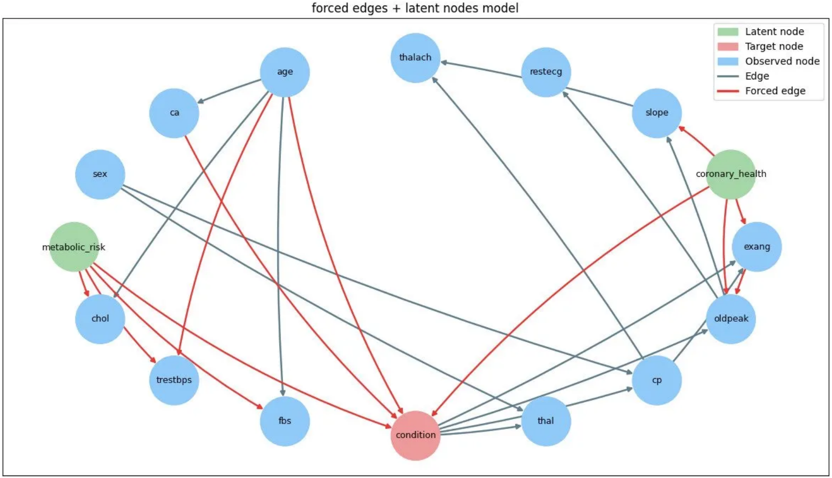
4

Latent Nodes

Two hidden variables: `coronary_health`, `metabolic_risk`. Parameters via EM.

Structure learning: **pgmpy** Hill Climbing, AIC scoring. Parameter estimation: Bayesian Estimator (BDeu prior). Inference: Variable Elimination

Forced Edges + Latent Nodes Model



Forced Edges: Medical Justification

Edge	Justification	*Source*
age → condition	Age is a primary non-modifiable risk factor for CAD	*Lima Dos Santos et al., 2023*
age → trestbps	Blood pressure systematically increases with age	*Lima Dos Santos et al., 2023*
ca → condition	Number of major vessels is a direct diagnostic marker for CAD severity	*AHA Guidelines*
oldpeak → condition	ST depression $\geq$1mm is the standard criterion for a positive stress test	*Katheria et al., 2021*
exang → oldpeak	Exercise-induced angina and ST depression co-occur in ischemia	*Clinical evidence*

Latent Variables: Clinical Motivation

Unobserved clinical constructs introduced to explain correlations among observed variables. Parameters estimated via Expectation-Maximization (EM).

`coronary_health`

Children nodes: `oldpeak`, `exang`, `slope`, `condition`

Clinical meaning: Represents inducible myocardial ischemia. ST depression, exercise-induced angina and ST slope are all manifestations of insufficient coronary perfusion under stress.

Source: StatPearls: Chest Pain clinical guidelines

`metabolic_risk`

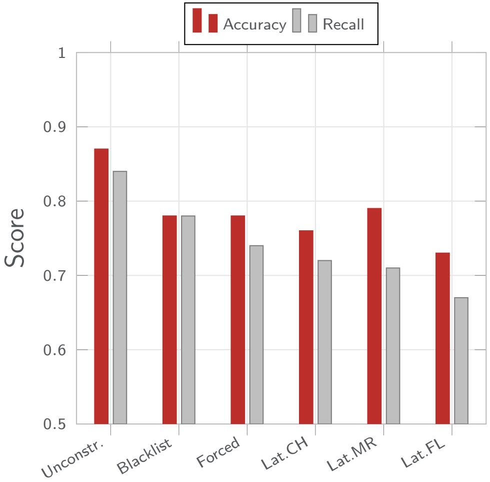
Children nodes: `chol`, `trestbps`, `fbs`

Clinical meaning: Represents the metabolic syndrome. Elevated cholesterol, blood pressure and fasting glucose are components of a single cardiometabolic risk cluster that jointly increases cardiovascular risk.

Source: Grundy, 2004, Metabolic Syndrome and CVD Risk

Results: Cleveland Dataset (5-fold Cross-Validation)

Model	Acc.	Recall
Unconstrained	**0.87**	**0.84**
Blacklist	0.78	0.78
Forced edges	0.78	0.74
Latent (cor. health)	0.76	0.72
Latent (metab. risk)	0.79	0.71
Latent (forced+latent)	0.73	0.67



The unconstrained model achieves the best accuracy. Domain knowledge improves **interpretability** but not **predictive performance**.

The Indian Heart Dataset model and Cross-Dataset Generalization

Experiment	Acc.	Rec.
Cleveland (unconstr.)	0.87	0.84
Clev. → Indian (latent)	0.54	0.57
Clev. → Indian (unconstr.)	0.68	0.80
Indian (own data)	0.81	0.75

Structural Invariance across Unconstrained Models

Similar edges found in both the unconstrained networks

Why does generalization fail?

- **Different distributions:** Indian population has distinct cholesterol and BP profiles
- **Demographics:** age distribution and sex ratio differ significantly
- **Missing feature:** Indian dataset lacks `thal`, weakening latent node
- **Small training set:** 303 patients → noisy CPDs, poor transfer
- Unconstrained transfers better (0.68 vs 0.54): latent nodes overfit to Cleveland-specific patterns

Conclusions

Main Findings

- **Unconstrained model (0.87)** achieved highest accuracy: data-driven learning is competitive against domain knowledge backed structures
- **Domain knowledge** improves interpretability but does not improve predictive performance
- **Latent variable models** underperformed, likely overfitting on small Cleveland dataset (303 patients)
- **Chest-pain features as universal predictors:** consistent causal parents across decades and geographies validate the extraction of robust medical knowledge
- **Cross-dataset generalization is partial**: the unconstrained model transfers reasonably (0.68 acc, 0.80 recall), the causal structure generalizes, but population-specific distributions affect the results

Limitations & Future Work

- **Dataset size:** 303 patients is small, learned CPDs are noisy, especially for latent variable models
- **AIC scoring** may favor overly complex structures: experiment with BIC for sparser networks
- **Future work:**
  - Larger, multi-center datasets
  - Richer latent variable structures
  - Transfer learning for cross-population robustness